

Pham Phuc Quang

AI ENGINEER

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CAREER OBJECTIVE

AI-Driven Software Engineer & Creative Technologist specializing in production-grade AI applications, multi-step Generative AI pipelines, and high-throughput interactive systems. Leveraging modern AI-first engineering workflows (Vibe Coding & Agentic Development with Antigravity and Cursor), I accelerate product delivery cycles by up to 5x while maintaining clean, scalable, and secure system architectures. Proven track record in building fact-checked RAG systems, automated AI video/audio generation pipelines, computer vision engines, real-time analytics dashboards, and multi-screen interactive systems for global enterprises. Seeking to tackle high-impact AI challenges by bridging robust backend engineering (Python & Node.js) with cutting-edge LLM and Machine Learning capabilities.

EDUCATION

2019 - 2026

Thuy Loi University

Bachelor of Engineering in Information Technology (Major: Information Technology)

• Graduation Thesis: “Applying Artificial Intelligence Solutions in Healthcare”

CERTIFICATIONS

2026

Completed Advanced University Training Program in Artificial Intelligence and Machine Learning.

WORK EXPERIENCE

06/2025 - Present

URAH NETWORK MEDIA JOINT STOCK COMPANY

AI-Driven Software Engineer (AI Engineer / Creative Technologist)

• Spearheaded AI-driven development workflows (Vibe Coding with AI Agents such as Antigravity & Cursor) to rapidly architect, build, and ship full-stack production AI systems at scale.

• Owned the end-to-end design, engineering, deployment, and monitoring of real-time interactive AI applications, high-throughput tracking pipelines, and analytics dashboards for multinational brands.

Key Production Projects Delivered:

1. Real-Time Multi-Screen Interactive Web System (GE Vernova Corporate Event)

• Project Overview: Built a spatial interactive web platform for GE Vernova’s flagship event, enabling real-time state synchronization between two independent web applications: a tablet/mobile Controller Web App and a main stage LED Display Web App.

• Technical Solution: Architected a low-latency bidirectional communication protocol using Node.js, Socket.io, and WebSockets, achieving sub-50ms synchronization latency. User gestures and inputs on the controller instantly streamed event payloads through the Node.js server to trigger complex GSAP and HTML5 Canvas animations on the main LED display.

• Key Achievements: Operated with 100% uptime and zero frame drops or disconnections throughout the live corporate event, delivering a seamless interactive experience for all

attendees.

2. Personalized AI Video Greeting Zalo Mini App & Real-Time Analytics Dashboard (Abbott Ensure Gold Vietnam)

• Project Overview: Developed a nationwide campaign platform on Zalo Mini App for Abbott Ensure Gold, enabling users to generate personalized greeting cards and AI-synthesized videos while providing real-time campaign telemetry for brand managers.

• High-Throughput Tracking Pipeline: Engineered a custom client-side telemetry SDK to capture granular user behavior (clickstreams, dwell time, step-by-step funnel conversion, and video completion rates), streaming events through load-optimized backend APIs to handle high-concurrency traffic spikes.

• Real-Time Analytics Dashboard: Built a centralized admin dashboard using React and Chart.js to visualize conversion funnels, live AI video render queue status, and real-time campaign KPIs.

• Key Achievements: Reliably processed over 50,000+ personalized AI video generations with 100% tracking data integrity, empowering the brand team to optimize campaign performance in real time.

3. AI Face-Swap Video Generation Web Platform (Nutifood)

• Project Overview: Built an interactive brand campaign web application for Nutifood where users upload portrait photos and an automated AI pipeline seamlessly swaps their faces into template promotional video clips.

• Technical Solution: Implemented a distributed, event-driven Task Queue using Redis and Node.js to buffer client requests and load-balance workloads across GPU-backed Python AI Face Swap worker servers. Optimized output video compression and integrated global CDN delivery for low-latency downloading and social sharing.

• Operations & Telemetry Dashboard: Built a real-time monitoring dashboard tracking GPU utilization, queue turnaround times, and user demographic analytics for executive reporting.

4. Real-Time Interactive Digital Human System (Global 'AI for Good 2026' Competition)

• Project Overview: Delivered an end-to-end conversational Digital Human solution showcasing next-generation human-AI interaction at the international 'AI for Good 2026' summit.

• Technical Solution: Orchestrated three core AI modules in a low-latency streaming pipeline: noise-resilient Speech-to-Text (STT) for live venue environments, context-aware conversational AI powered by OpenAI GPT-4 API, and real-time Text-to-Speech (TTS) synchronized with viseme-based facial lip-sync.

• Key Achievements: Achieved natural, human-like conversational interactions with under 2 seconds of end-to-end response latency, successfully representing the team on an international stage.

5. AI Virtual Historical Ambassador – RAG Q&A System (National Exhibition)

• Project Overview: Deployed a specialized domain-specific AI Q&A Assistant serving visitors at the National Exhibition '80 Years of Journey to Independence - Freedom - Happiness'.

• Technical Solution: Architected a standardized historical knowledge base using Retrieval-Augmented Generation (RAG) and vector search, enforcing strict grounding to guarantee 100% factual accuracy on sensitive historical topics. Implemented multi-layer content moderation guardrails and optimized backend throughput for high visitor concurrency.

6. Interactive GenAI Campaign Ecosystem: 'Rising from Heritage' – AI Magazine & AI Music Game (MSB Bank)

• Project Overview: Built an interactive web ecosystem for two flagship AI campaigns of Maritime Bank (MSB): 'Future Heritage Magazine Cover' (personalized GenAI magazine portraits) and 'MSB AI Music Game – With Quang Hung MasterD' (LLM lyric composition, AI vocal synthesis, and automated MV rendering).

• Technical Solution: Engineered a Multi-SPA frontend (React, TypeScript, Vite, Tailwind CSS v4) deployed on Firebase Hosting & Global CDN. Built a multi-step AI backend pipeline integrating LLM lyric generation, AI voice/music synthesis aligned with backing tracks, multi-

layer Brand Safety moderation, automated lyric video rendering, and TikTok engagement tracking.

• Key Achievements: Served 10,500+ active participants, generating 2,729 AI magazine covers and 4,068 AI songs/music videos, driving 257 TikTok contest entries with over 3.59 million total engagements.

7. Personalized AI Voice & Video Generation Platform – 'Cheer to the End with MILO' (Nestlé Vietnam)

• Project Overview: Built an interactive web platform for Nestlé Vietnam enabling users to upload portrait photos and select themes to automatically generate personalized AI cheering voiceovers and face-tracked Full HD karaoke/cheering videos.

• Technical Solution: Developed a React 19 + TypeScript frontend and a hybrid Backend Media Engine (Node.js Express orchestrating multi-threaded Python, OpenCV, MediaPipe, PyTorch, and FFmpeg workers) deployed on Google Cloud VMs & Platform.sh. Integrated ElevenLabs TTS (eleven_v3), HMAC-SHA256 signed Zalo ZNS/Voice OTP verification, and computer-vision + LLM Brand Safety guardrails.

• Key Achievements: Passed 100% of Nestlé's rigorous enterprise security audits (OWASP ASVS Level 2 & OWASP API Security Top 10), delivering high-speed multi-threaded Full HD video rendering via Firebase Storage.

8. Large-Scale QR Interactive Platform 'Nutifood Pedestrian Bridge' – Coordinate Map & GenAI Media (Nutifood 2026)

• Project Overview: Engineered a massive-scale interactive web platform for Nutifood's Pedestrian Bridge campaign, where users scan unique QR codes and verify OTPs to claim personal coordinates on the bridge map and generate personalized AI ID cards and videos.

• Technical Solution: Architected a containerized Backend (Node.js, Express, TypeScript, Docker on Linux GCP VM) optimizing a 29-million-record SQLite QR database (3GB) to achieve sub-200ms query latency. Built a stateful QR Resume Flow, automated image/text moderation using Google Gemini AI, a bridge coordinate pinning algorithm, and a 2FA-secured Admin Dashboard (Argon2id + TOTP).

• Key Achievements: Guaranteed 100% seamless session recovery across 29 million unique QR codes with instant lookup latency and comprehensive real-time operational reporting and audit logging.

05/2025 - 06/2025 **TriAn Joint Stock Company**

Junior AI Engineer

• Developed AI-powered mobile application features for TriAn to assist elderly users with daily health tracking and stress-relief conversational support.

• Built high-performance backend APIs integrating Small Language Models (SLMs) with the company's core CRM system.

07/2024 - 05/2025 **URAH NETWORK MEDIA JOINT STOCK COMPANY**

Machine Learning Intern

• Contributed to developing the AI-integrated TriAn mobile application for elderly health monitoring and empathetic conversational AI.

• Engineered low-latency APIs connecting Small Language Models (SLMs) with core CRM workflows.

• Researched and applied Model Quantization and optimization techniques to deploy AI models efficiently on resource-constrained edge hardware, including NVIDIA Jetson Nano and GTX 1050Ti workstations.

TECHNICAL SKILLS

AI-Driven Engineering	AI-First Development (Vibe Coding), Advanced Prompt Engineering, Structured Outputs, Function Calling, AI Coding Agents (Antigravity, Cursor, Claude Code), System Architecture

& Agentic Workflows	Design.
AI, Machine Learning, LLM & Computer Vision	LLM APIs (OpenAI GPT-4, Google Gemini, Claude), RAG Systems & Vector Search, Computer Vision (Python, PyTorch, OpenCV, MediaPipe, Face Swap), TTS/STT (ElevenLabs), TensorFlow, Keras, Model Quantization, Classical ML (scikit-learn, KNN, SVM, Decision Trees).
Backend & Full-Stack Engineering	Python, Node.js, Express, TypeScript, JavaScript (ES6+), React, RESTful APIs, Redis Task Queues, Docker, Linux VM, SQLite/SQL, Firebase Ecosystem (Firestore, Realtime DB, Storage, Cloud Functions, Auth), OWASP Security.
Real-Time Systems & Edge AI Hardware	WebSockets, Socket.io, Sub-50ms Multi-Screen State Synchronization, FFmpeg Multi-Threaded Media Pipelines, Edge AI Deployment (NVIDIA Jetson Nano, GPU GTX 1050Ti).
Data Pipelines, Analytics & Observability	High-Throughput Event Tracking SDKs, Real-Time Analytics Dashboards (React, Chart.js, Recharts), Python Data Stack (NumPy, pandas, matplotlib), Git, Azure ML, Google Cloud Platform (GCP).

INTERESTS

Reading technical literature, Running, Traveling

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